

NGS-DOSE on the 1000 Genomes 30× cohort: a trio-focused assessment

as of 2026-09-23, from the tables `ngsdose` report writes; nothing is typed in

The question

Can the copy number of a sequence that a reference genome collapses — the ribosomal DNA arrays, several hundred copies per genome — be measured from ordinary short-read whole-genome sequencing? No assay of rDNA copy number exists for these samples, so the answer has to come from things that are known: sequence of known copy number in every genome, Mendelian transmission in trios, the same genome counted two ways, and long-read assemblies of the same people for the satellite arrays that the same machinery measures.

What is measured

Reads are assigned to a class (45S rDNA, 5S rDNA, the distal junction, ten satellite families, the telomeric repeat) by 31-mers that occur in the class and nowhere else in GRCh38 or CHM13; where the aligner put a read is not used. Fragment 5' ends are counted on the unit and in 800 single-copy control regions. A per-library Poisson spline of fragment-end density on fragment GC content, fitted on the controls, gives the expected count of any sequence; copy number is $2 \times \text{observed} / \text{expected}$ per 250-bp window, and the windows of the 45S unit are calibrated across the cohort with the scale set by anchor windows on which three Illumina chemistries agreed. Two counting modes: scan reads the whole CRAM, fetch retrieves only the controls and the intervals where the aligner places class reads. Cohort: the 1000 Genomes 30× CRAMs (NYGC, NovaSeq 2×150, GRCh38), all lymphoblastoid cell lines.

Three 45S estimators travel through every table, two of them NGS-DOSE's. '45S, NGS-DOSE calibrated': the cohort model $\log C(i,w) = c(i) + a(w) + e(i,w)$ over every retained 250-bp window w of the unit, fitted by median polish across samples i with the window efficiencies $a(w)$ pinned to a median of zero over the anchor windows; the estimate is $\exp(c(i))$. '45S, NGS-DOSE single-sample': $2 \times \text{observed} / \text{expected}$ fragment ends over the anchor windows alone, under the sample's own GC model, with no information from any other sample. '45S, 18S depth ratio (published)': $2 \times \text{fragment ends in the 18S gene} / (\text{positions} \times \text{the control regions' mean rate})$, no GC model, no calibration — the estimator of published studies, computed from the same reads for comparison; it is not NGS-DOSE's estimate.

How it is tested

In a trio, a child's copy number is the mean of the parents' plus segregation; measurement error is not inherited. Values are centred within population; the correlations are Pearson's. The child-midparent correlation cannot reach 1 even for a perfectly measured heritable trait, because half of a child's variance is segregation, which the midparent does not predict (r is bounded by about $\sqrt{((1 + \rho)/2)}$, 0.71 at $\rho = 0$). The least-squares slope b of child on midparent has no such ceiling, so the slope, corrected for the spousal correlation ρ , is the statistic that answers what share of the measured variance is real: reliability $R = b - \rho(1 - b)$, 1 for a perfectly measured heritable trait and 0 for pure error, with family-bootstrap 95% intervals and a one-sided p -value for the slope from 1,000 permutations of children among families. The test is applied to every metric in four groups. The rDNA estimators are the claim. The satellite arrays are positive controls: their mass is a property of the genome, measured by the same k -mer machinery, and must be inherited. Known copy numbers (held-out autosomal sequence, the distal junction) have nothing to inherit. The culture's and the library's properties (mitochondrial and EBV content, depth, duplicate rate, GC bias, insert size) are not in the nuclear genome. A method that measured library artefacts would light up the last group; one that measured nothing would light up none. The same test is then run on the fetch counts alone, and the satellite masses are held against HPRC assemblies of the same people.

Results

735 of 3,202 genomes, 149 complete trios, as of 2026-09-23. Known copy numbers: held-out autosomal sequence 1.999 ± 0.008 (expected 2); the distal junction 9.67 ± 0.34 (expected 10); sex from the reads agrees with the pedigree in 735 of 735. Transmission, 45S calibrated estimate: reliability 1.00 (0.93–1.21), child-midparent r 0.78, spousal r 0.17. 5S: 0.78 (0.46–1.11). Positive controls, 9 satellite families: reliability 0.43 to 1.00 (HSat1B, on Yq, excluded: father-to-son only); the telomeric repeat, which changes with age and in culture, 0.33. Negative controls: culture and library -0.27 to 0.06; known copy numbers -0.07 to 0.69. Fetch counts alone: 45S reliability 1.00; fetch/scan agreement of the 45S estimate across genomes r 1.0000. Targeted fetch against whole-file scan, 45S: median 0.9997 (0.9992–0.9999) over 735 genomes. Parents and children were sequenced apart: 149 of 149 children in release batch 698, 287 of 298 parents in 2,504. No batch is shared within a family, but one that reads a generation on another scale moves R with it: HSat2 array spreads 1.15× as much in the children as in their parents and reads R 1.23, or 1.07 with the children on their parents' scale; ACRO array spreads 0.86× as much in the children as in their parents and reads R 0.86, or 0.99 with the children on their parents' scale. Assemblies: 2 of 10 satellite families track HPRC assemblies of 54 of these genomes with $r \geq 0.95$ (HSat1B, ACRO).

Limits

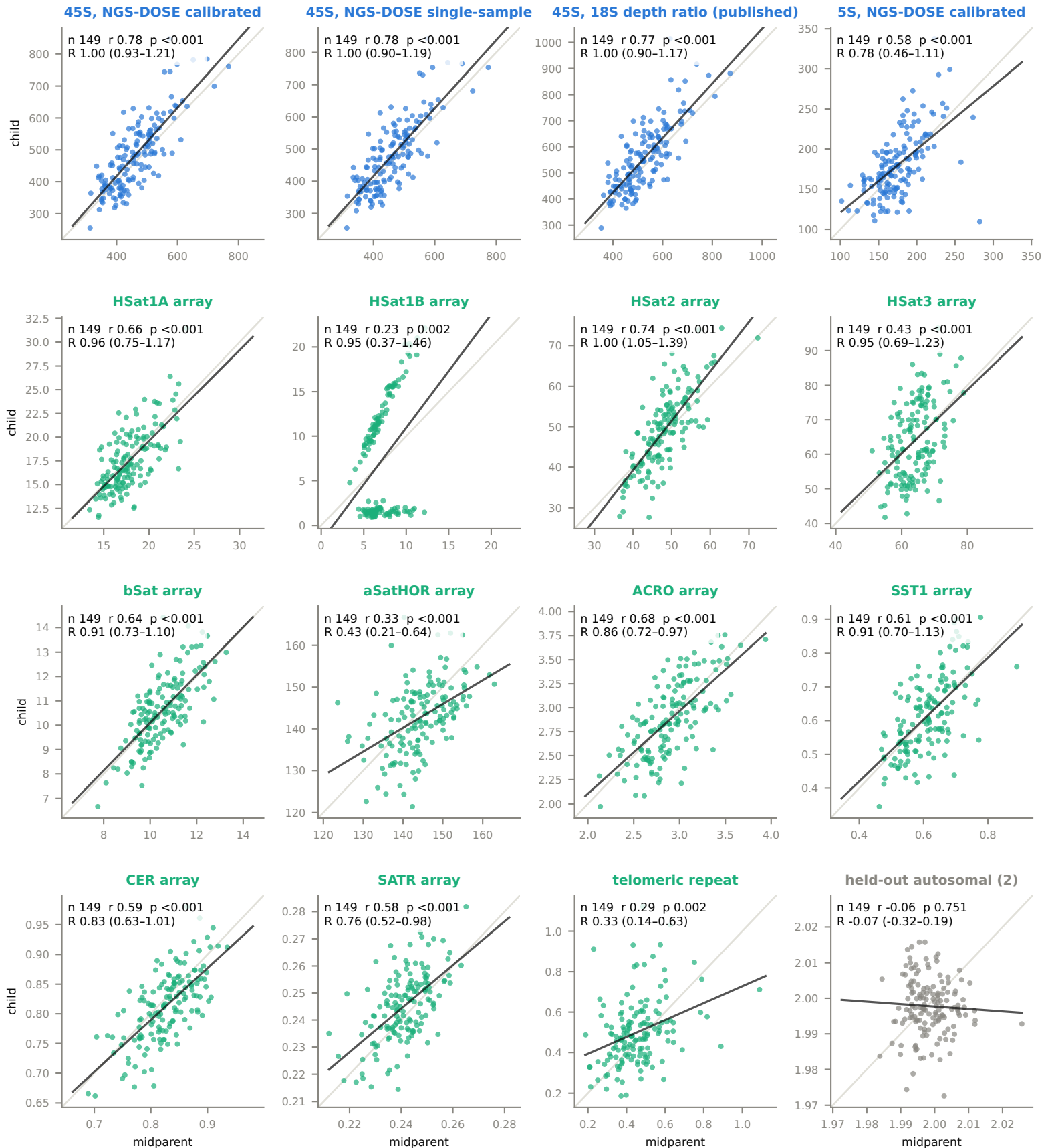
What this does not show: an absolute calibration of rDNA copy number (none exists for these samples; the scale rests on unit windows on which three chemistries agree); anything about DNA that is not from a lymphoblastoid cell line or a chemistry other than NovaSeq 2×150; and, because people differ in 45S copy number far more than any competent estimator errs, the trios show that the measured variation is inherited, not which rDNA estimator is best — that ranking rests on the same people sequenced on two technologies (the pilot), where the calibrated estimate reproduced and the depth ratio did not.

Transmission of every metric, at a glance

		child-father r	child-mother r	child-midparent r	spousal r	midparent slope	reliability R	R from fetch
rDNA copy number (the claim)	45S, NGS-DOSE calibrated	0.62	0.57	0.78	0.17	1.05	1.06	1.06
	45S, NGS-DOSE single-sample	0.62	0.57	0.78	0.17	1.03	1.04	1.04
	45S, 18S depth ratio (published)	0.60	0.58	0.77	0.19	1.02	1.03	1.03
	5S, NGS-DOSE calibrated	0.46	0.37	0.58	0.03	0.79	0.78	0.78
satellite arrays, mass (genomic: positive controls)	HSat1A array	0.49	0.43	0.66	-0.01	0.96	0.96	
	HSat1B array	0.25	-0.03	0.23	-0.23	0.94	0.95	
	HSat2 array	0.52	0.52	0.74	-0.02	1.23	1.23	
	HSat3 array	0.39	0.16	0.43	-0.11	0.95	0.95	
	bSat array	0.43	0.48	0.64	0.02	0.91	0.91	
	aSatHOR array	0.28	0.20	0.33	0.01	0.43	0.43	
	ACRO array	0.44	0.50	0.68	-0.05	0.85	0.86	
	SST1 array	0.44	0.42	0.61	0.02	0.91	0.91	
	CER array	0.40	0.45	0.59	0.04	0.83	0.83	
	SATR array	0.41	0.46	0.58	0.14	0.79	0.76	
	telomeric repeat	0.19	0.23	0.29	0.08	0.38	0.33	
known copy number (nothing to inherit except the DJ's whole-copy steps)	held-out autosomal (2)	-0.07	-0.02	-0.06	-0.01	-0.08	-0.07	-0.07
	distal junction (10)	0.43	0.26	0.50	-0.06	0.67	0.69	0.69
	mitochondrial genomes per cell	0.07	-0.01	0.04	-0.00	0.05	0.06	0.06
culture and library (not in the nuclear genome)	EBV episomes per cell	0.00	0.16	0.09	0.13	0.12	0.01	0.01
	sequencing depth	-0.06	0.02	-0.02	-0.05	-0.01	0.04	0.04
	duplicate-flagged fraction	-0.03	0.08	0.03	0.07	0.05	-0.02	-0.02
	library GC bias	-0.06	0.03	-0.03	0.06	-0.04	-0.10	-0.10
	insert size	-0.16	-0.09	-0.17	0.09	-0.16	-0.27	-0.27

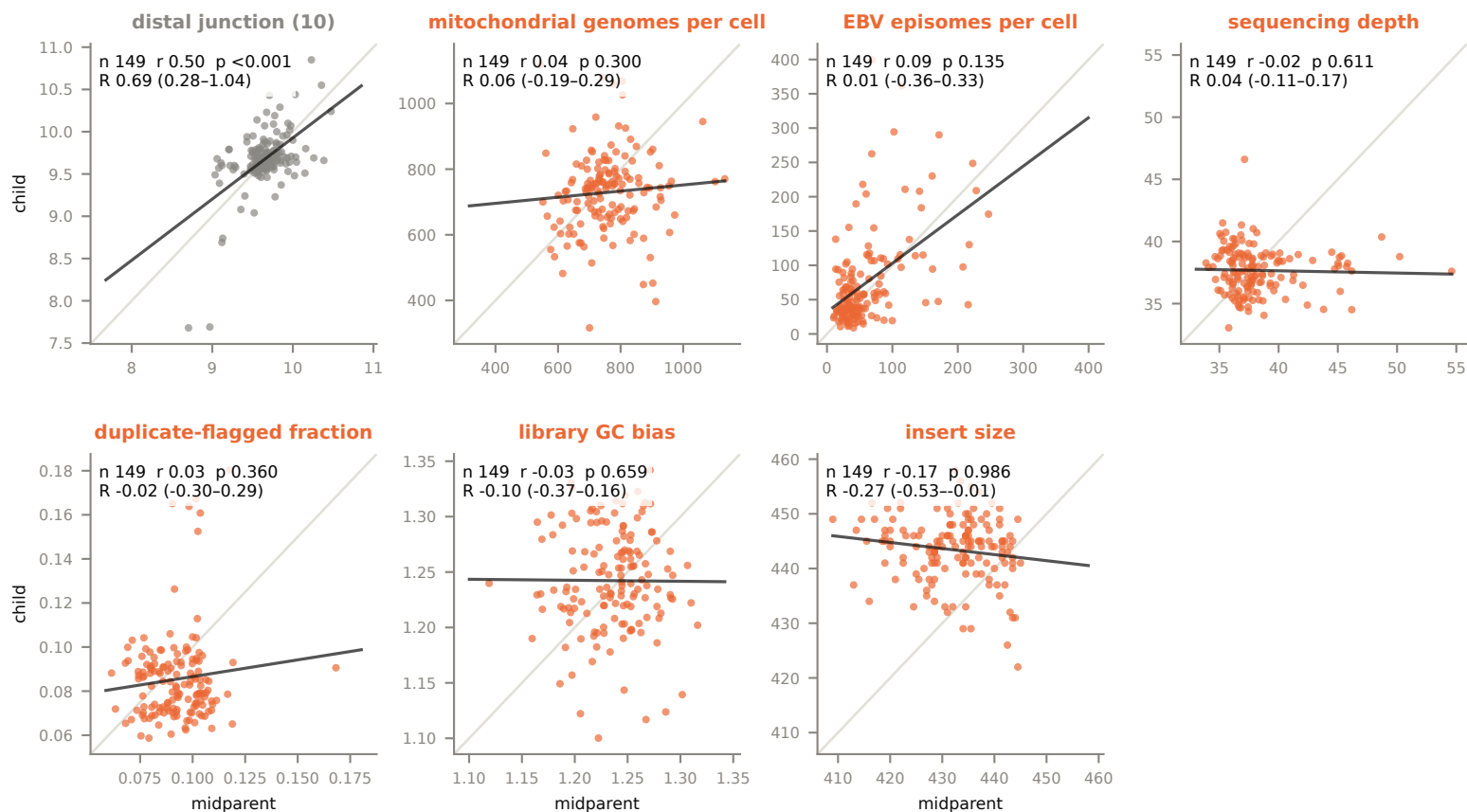
Colour and number: the statistic, from 0 (white) to 1 (blue); values are within population, over the complete trios. A metric that is inherited reads near 1 in the correlation columns and in R; a property of the culture or the library reads near 0; sequence of known copy number has no variation to inherit and reads near 0 too. HSat1B lives mostly on Yq and passes from father to son only, which dilutes its midparent statistics by design. The last column repeats R with the cohort layer and the trio test run on the fetch counts alone.

Child against midparent, every metric



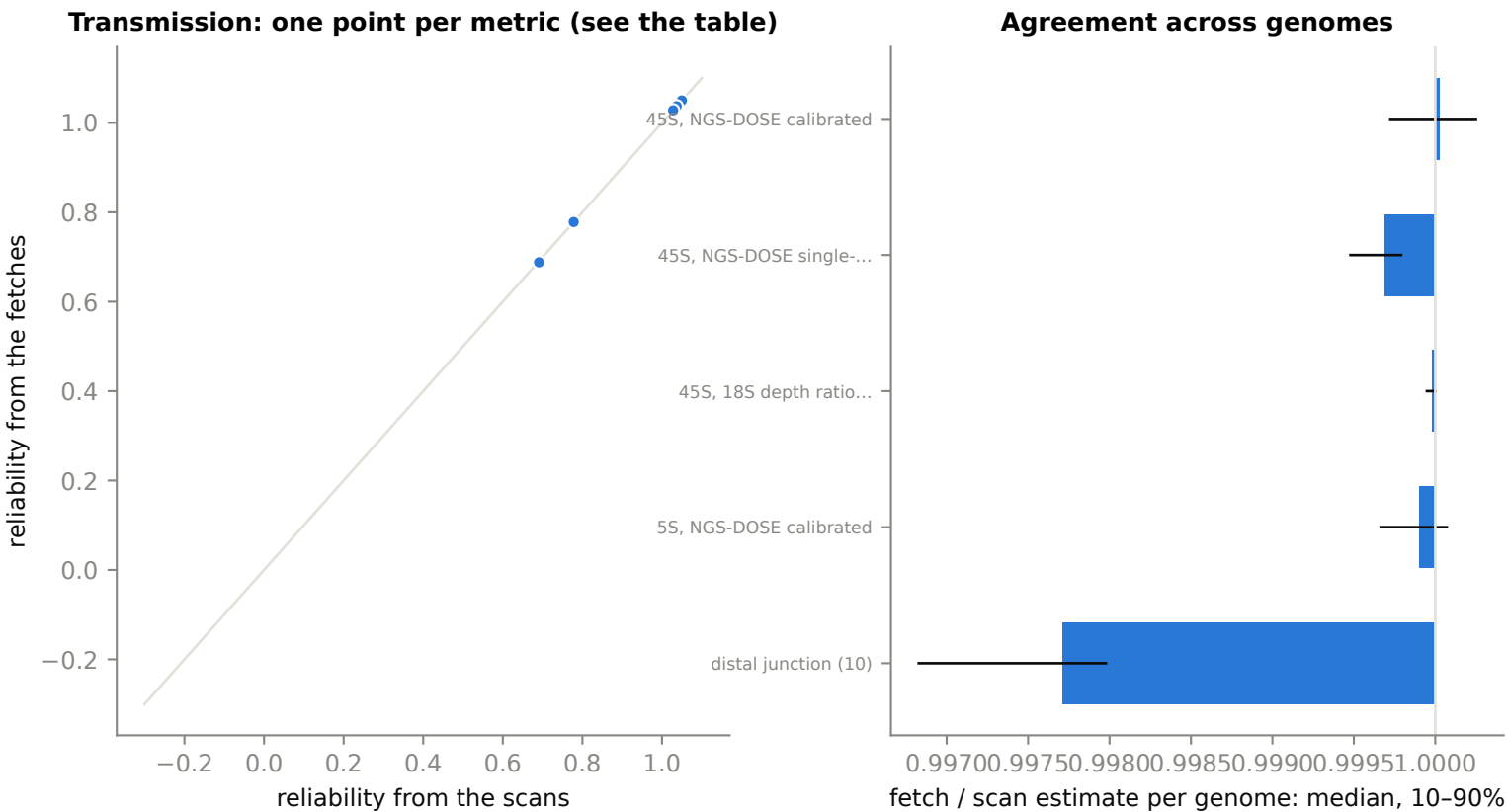
Raw values (not centred); grey diagonal: child equals midparent; black line: least squares. Printed: the report's statistics, computed within population (n trios, Pearson child-midparent r, the permutation p of the midparent slope, reliability R with its family-bootstrap 95% interval). Colours: blue rDNA estimators, green satellite arrays (positive controls), grey known copy numbers, orange culture and library (negative controls).

Child against midparent, every metric (continued)



Raw values (not centred); grey diagonal: child equals midparent; black line: least squares. Printed: the report's statistics, computed within population (n trios, Pearson child-midparent r, the permutation p of the midparent slope, reliability R with its family-bootstrap 95% interval). Colours: blue rDNA estimators, green satellite arrays (positive controls), grey known copy numbers, orange culture and library (negative controls).

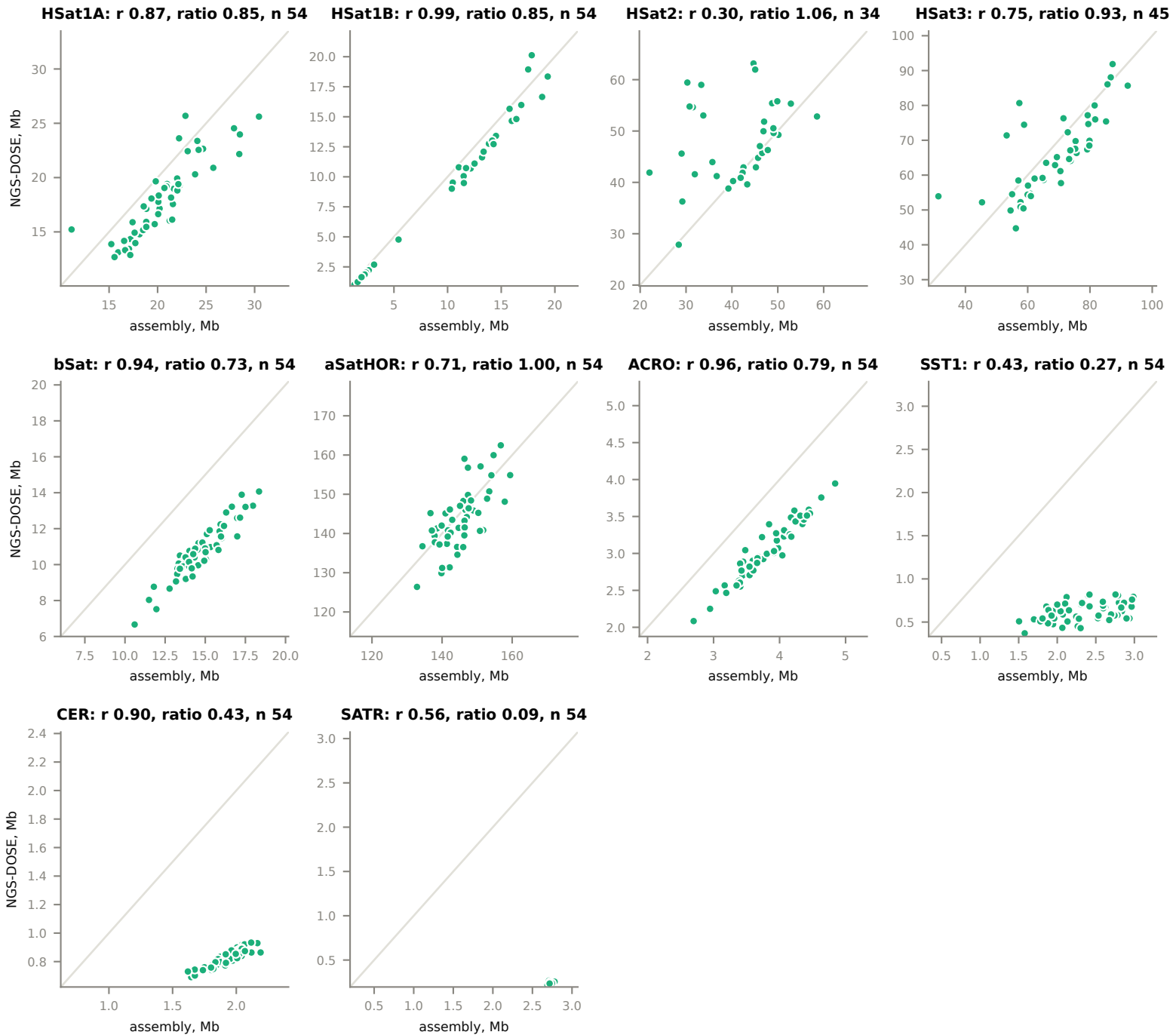
The same test on the fetch counts alone



metric	genomes	fetch / scan	r across genomes	R from scan (95% CI)	R from fetch (95% CI)
45S, NGS-DOSE calibrated	735	1.0000	1.0000	1.00 (0.93-1.21)	1.00 (0.93-1.20)
45S, NGS-DOSE single-sample	735	0.9997	1.0000	1.00 (0.90-1.19)	1.00 (0.90-1.19)
45S, 18S depth ratio (published)	735	1.0000	1.0000	1.00 (0.90-1.17)	1.00 (0.90-1.17)
5S, NGS-DOSE calibrated	735	0.9999	1.0000	0.78 (0.46-1.11)	0.78 (0.46-1.11)
distal junction (10)	735	0.9977	0.9998	0.69 (0.28-1.04)	0.69 (0.28-1.04)

The cohort layer (window calibration, control-region PCs) and the trio test were run on the fetch counts of 735 genomes without reference to the scans. Columns made from the same reads in both modes come out identical and are omitted (held-out autosomal (2), mitochondrial genomes per cell, EBV episomes per cell, sequencing depth, duplicate-flagged fraction, library GC bias, insert size, chrX, chrY). The classes differ by what the sinks miss and by a calibration learned twice.

Satellite arrays against long-read assemblies of the same people



54 of the genomes have HPRC release-2 assemblies; the CenSat annotation of both haplotypes gives the size of every satellite array, the same kind of sequence the k-mer machinery measures. A genome is compared in a class only when the arrays its assembly did not close are immaterial. Grey diagonal: equality. The relative measures (β -satellite, CER, ACRO) sit below it by a constant factor, their k-mer recall. Tracking the assemblies across people with $r \geq 0.95$: HSat1B, ACRO. Assemblies collapse the rDNA itself and are no truth for it.

Sequence of known copy number, and fetch against scan

what	expected	measured	n
held-out autosomal sequence (80 regions)	2	1.999 ± 0.008	735
chrX in men (60 regions)	1	0.991 ± 0.006	350
chrX in women with an intact culture	2	1.933 ± 0.025	370
chrY in men with an intact Y (40 regions)	1	0.975 ± 0.014	343
chrY in women	0	at most 0.0066	385
distal junction (one per acrocentric short arm)	10	9.67 ± 0.34	735
sex inferred from the reads	pedigree	agrees in 735 of 735	
fetch / scan, 45S	1	0.9997 (0.9992–0.9999)	735
fetch / scan, 5S	1	0.9999 (0.9994–1.0000)	735
fetch / scan, distal junction	1	0.9978 (0.9967–0.9983)	735
fetch / scan, held-out autosomal	1	1.0000 (1.0000–1.0000)	735
fetch / scan, chrM	1	1.0000 (1.0000–1.0000)	735

Every genome carries sequence whose copy number is not in question, measured by exactly the code that measures the rDNA: held-out autosomal regions, the X and Y, and the distal junction, a 400-kb sequence present once on each of the ten acrocentric short arms — multi-copy, paralogous and acrocentric, the kind of sequence the rDNA is. Women read the X and every genome reads the distal junction a few percent below expectation; both are late-replicating sequence, which DNA from a growing culture under-represents.

Relative to the cohort's level the distal junction sits at whole numbers: 4 genomes at -2 , 16 at -1 , 614 at 0, 21 at $+1$ (robust SD 0.15 copies). A step is a structural variant of an acrocentric short arm; where a carrier parent and a child were both counted it was transmitted in 7 of 22, and 1 child(ren) carry a step neither counted parent has.